

Frontend Development Overview

Frontend development involves building the user-facing part of web applications. The core technologies include HTML, CSS, and JavaScript.

1. HTML (HyperText Markup Language)

Defines the structure and content of a webpage.

2. CSS (Cascading Style Sheets)

Styles and layouts for web pages, including Flexbox, Grid, animations, etc.

3. JavaScript

Adds interactivity and dynamic behavior to web pages.

4. Frontend Frameworks and Libraries

- React: Component-based library from Facebook.
- Vue.js: Progressive and approachable framework.
- Angular: Full-fledged framework from Google.

5. Build Tools

- Webpack: Module bundler.
- Vite: Fast build tool optimized for modern development.
- Babel: JavaScript compiler for backward compatibility.

6. State Management

- Vuex / Pinia for Vue.
- Redux for React.

7. Testing Tools

- Unit Testing: Jest, Mocha
- E2E Testing: Cypress, Playwright

8. Deployment

- Methods include Nginx, CI/CD pipelines, and CDN distribution.

Staying up to date and continuous practice are key to becoming a skilled frontend developer.